G1JKS QMX Mini-Microphone

Setup and Tweaking guide

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Like many amateur radio microphones, inside the G1JKS QMX mini-microphone is a Electret condenser microphone module than needs a phantom DC bias voltage to power the internal preamp FET, and a 2k2 load resistor. The phantom bias voltage is delivered from the QMX radio through the TRS cable, with the 2k2 load resistor radio side of the cable (i.e. on the QMX PCB). This is different to some other radios (e.g. the KX2 and KX3) where the 2k2 load resistor (required by the Electret FET to function correctly) resides in the microphone itself.

For anyone interested in learning more about the electronics inside the QMX radio, please scan the QR code below to bring up the latest schematics (version 3). On page two you will find X202 and R218 on the right-hand side of the schematic; X202 is the “Paddle” port into which the microphone TRS cable should be plugged when operating in SSB mode.

The Electret microphone module inside the G1JKS Mini-Microphone is a very sensitive PUI Audio AOM-5024P-HD-MB-R, at -24dB. This guide has therefore been created to help you tweak the settings on your QMX radio to set the microphone audio level to your personal preference with this particular Electret module.

1. Running the build-in “Microphone test”

One of the many nice features of the QMX radio is it has a build-in microphone test that allows to you determine what audio gain level to set in the “SSB” configuration menu. To run the microphone test, begin by pressing and holding the little “Menu” button as shown below until “Audio” is displayed:

Next rotate the right-hand “Tune” knob until “Hardware tests” is displayed, then press the “Menu” button once again so “AF filter sweep” is displayed.

Next rotate the “Tune” knob until “Microphone test” is displayed, and then press the “Menu” button once again to enabled the microphone test:

Now plug in the G1JKS QMX mini-microphone into one end of the supplied TRS cable, and the other end in to the “Paddle” port of your QMX radio. You should now observe the top bar graphs changing as you speak into the microphone (note you do not need to press the PTT button to do this test). Next using the right-hand “Tune” knob adjust the audio gain ensuring that the top bar-graph does not saturate.

Once you have decided on the right gain level for your voice, move on to the next section to learn how to set it permanently in the “SSB” configuration menu,

2. Making the Microphone gain level persistent

Once you have run the microphone test described in section 1 above, you need to set the desired audio gain level so that it is persistent. One does this through the “SSB” configuration menu described below.

First press and hold the “Menu” button, then rotate the right-hand “Tune” knob, until “SSB” is displayed:

With “SSB” displayed, press the “Menu” button briefly then rotate the right-hand “Tune” knob, until “Mic Gain” is displayed. Press the “Menu” button once again, then rotate the “Tune” knob until the desired gain level is set to your preference (from the Microphone test procedure described above).

Because of the sensitivity of the PUI Audio AOM-5024P-HD-MB-R, I find a low gain is absolutely fine for me, even when speaking into the microphone at a distance of around 15cm. Adjust for you own preferences.

To view a video of the above procedure by Andy (VK5LA), please Scan the QR code above..

73 Dave G1JKS

For more pictures, including the the colours available, please scan the following QR code: